

Year 4 Science Week 2026: Think, Test & Discover

Seeds of Science — APA National Science Week 2026 Learning Resource



What Makes Someone a Scientist?

Ask Questions

Wonder about the world around you

Predict

Make educated guesses based on what you know

Investigate

Test your ideas to find real answers

Collect Evidence

Record observations carefully and honestly

Find Patterns

Look for what your findings tell you

Explain

Share what you discovered and why it matters

 *Your challenge: If you could investigate one mystery about the world, what would it be?*

The Science Investigation Cycle



Question

Predict

Plan

Test

Example in action: Does a heavier ball roll faster than a lighter ball down a ramp? Scientists use the **same ramp, same height, and 3 repeated tests** — then explain using **evidence, not guesses!**

Forces: Push, Pull & Motion

A **force** is a push or pull that can change how something moves.

→ Kicking a football

Your foot pushes the ball, making it move

→ Opening a drawer

You pull the handle to slide it open

→ Pushing a trolley

You push to make it move faster or slower

→ Pulling a wagon

You pull the rope to move it along



📄 *Prediction challenge: Which object would be easiest to move – a tennis ball, a textbook, or a desk? What evidence could you collect to test your idea?*

Material Detectives

Different materials have different **properties** that make them useful for different jobs.



Strength

Can it hold weight without breaking?



Flexibility

Can it bend without snapping?



Hardness

Is it difficult to scratch or dent?



Absorbency

Does it soak up water?

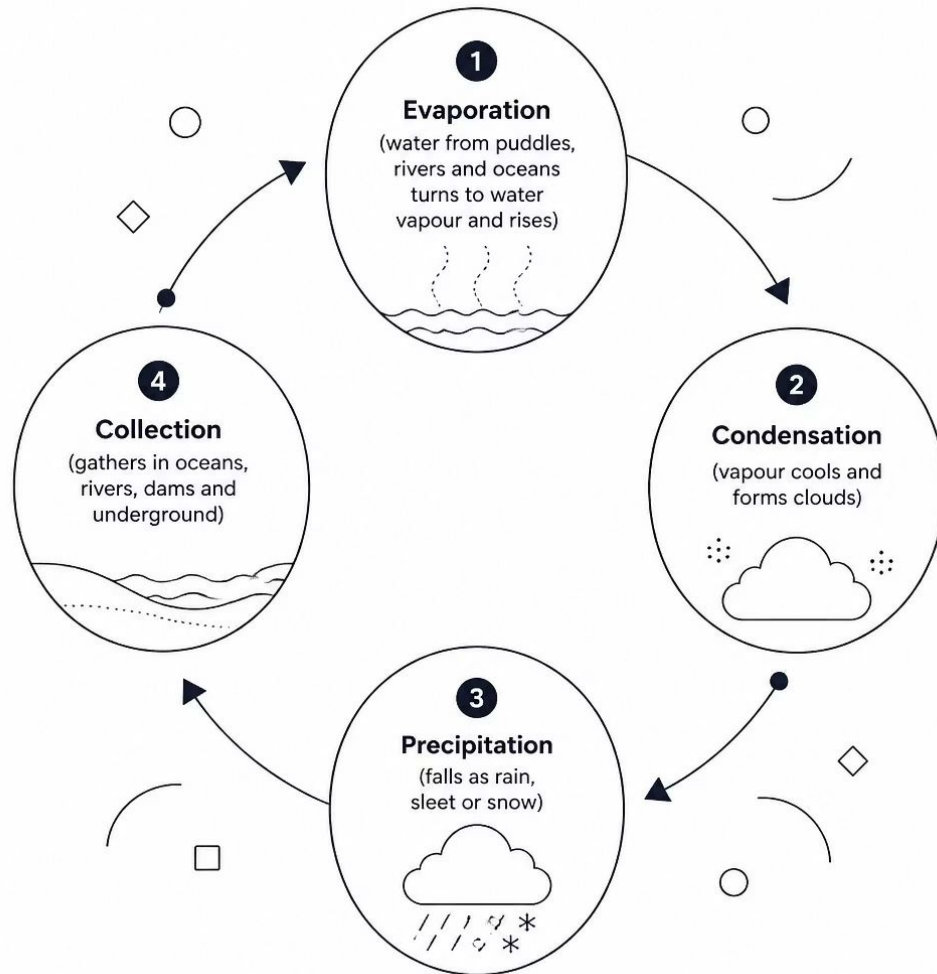


Waterproofing

Does water slide off it?

 *Your challenge: Which material would you choose to make a raincoat — cotton, plastic, or wool? Use what you know about material properties to explain your choice.*

Water Cycle Investigation



THE WATER CYCLE

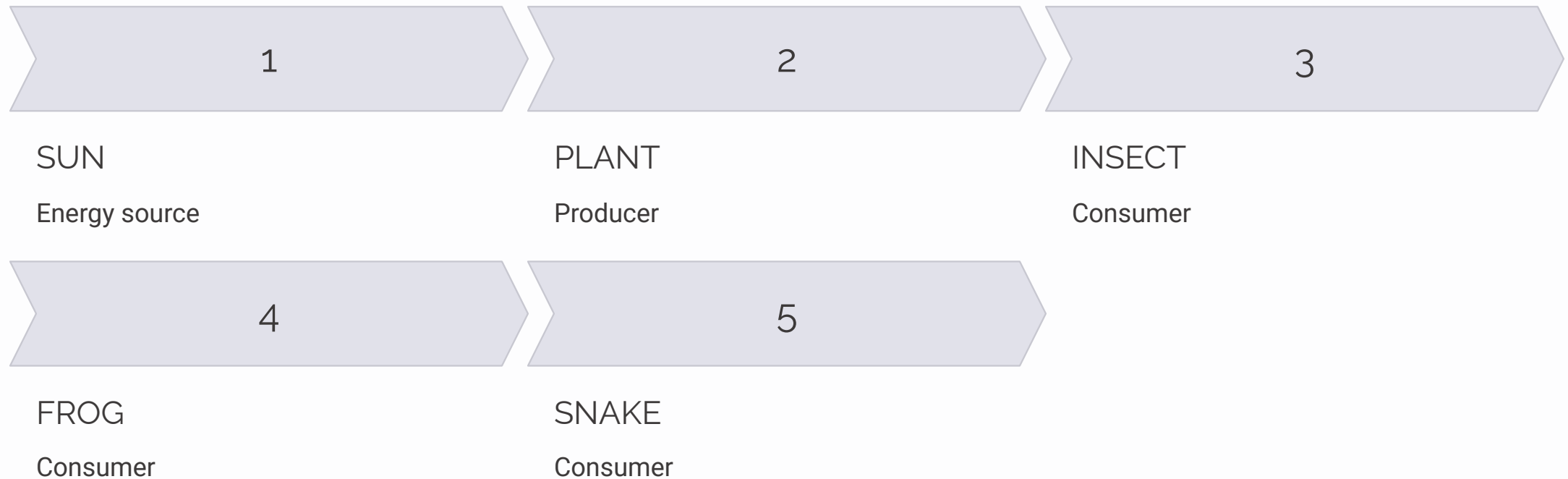
EVAPORATION →
CONDENSATION →
PRECIPITATION →
COLLECTION

- **Evaporation:** Water turns into invisible water vapour and rises into the sky
- **Condensation:** Vapour cools and forms tiny droplets – clouds!
- **Precipitation:** Water falls as rain, sleet or snow
- **Collection:** Water gathers in oceans, rivers, dams and underground

📄 *Think about this: Where do you think the water in a puddle goes after a sunny day?*

Who Eats Whom?

Every living thing has a role in its habitat. **Food chains** show who eats what.



- **Producers:** Plants make their own food using sunlight
- **Consumers:** Animals eat plants or other animals
- **Decomposers:** Fungi and bacteria break down dead things and return nutrients to soil

📄 *Reasoning challenge: What could happen to the food chain if all the frogs disappeared? Why?*

Mini Science Investigation: Water-Stopping Materials

Question

Which material is best at stopping water from getting through?

Materials to Test

Paper · Fabric · Foil · Plastic · Cotton wool

Fair Test — Keep These the Same

- Same amount of water for each material
- Same time to test (30 seconds)
- Same thickness of material
- Same temperature of water

Observe & Record

- Does water soak through?
- How long does it take?
- How wet does the material get?

Your conclusion: Which material was best? What evidence supports your answer?



Science Week Challenge Questions

Use your scientific thinking to answer these — explain your reasoning!

1

Forces

A toy car rolls down a steep ramp faster than a gentle ramp. Why? What force is involved?

2

Materials

A sponge absorbs water but a plastic bag does not. How could you use this difference to choose the right material for a job?

3

Water Cycle

If there were no evaporation, what would happen to the water in the ocean? Explain your thinking.

4

Food Chains

Plants are called producers. Why do you think they have this name?

5

Investigation

You want to test whether different coloured paper dries at different speeds. What would you need to keep the same to make it a fair test?

Teacher note: Answers available in speaker notes

Your Science Mission

Become a young scientist. Follow these steps:

01

Observe

Find something interesting around you

03

Predict

Make a thoughtful prediction

05

Collect

Gather your evidence

02

Question

Ask a science question about it

04

Plan

Design a fair test

06

Explain

Share what you discovered

Science begins with curiosity. Keep exploring with free Year 4